Red Hat Enterprise Linux 9.5

Bare Metal Deployment & Remote Management Standard Operating Procedure (SOP)

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1. Prerequisites & Preparation

Hardware Requirements

- **Architecture:** 64-bit x86 (x86_64) or ARM64.
- **Memory:** Minimum 8GB RAM (16GB+ recommended for production workloads).
- **Storage:** Minimum 50GB local disk or SAN-attached LUN.

Access & Network Requirements

- **Out-of-Band (OOB) Access:** IP, Username, and Password for the Baseboard Management Controller (BMC) - e.g., Dell iDRAC, HPE iLO, Lenovo XClarity, or generic IPMI.
- **Media:** The official rhel-baseos-9.5-x86_64-dvd.iso downloaded directly from the Red Hat Customer Portal.
- **VLAN Configuration:** Ensure switch ports for primary NICs are correctly tagged for the provisioning and production networks.

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2. Architecture Diagram

The following diagram illustrates the out-of-band management workflow used for remote bare-metal provisioning:

[PASTE IMAGE BLOCK HERE: Architecture diagram showing Administrator Workstation -> Management Network (VLAN) -> Server BMC/iDRAC/iLO port -> Internal Server Hardware (CPU/Memory/Disk)]

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3. Phase 1: Accessing Remote Console

1. Open a supported, secure web browser and navigate to the IP address of the server's BMC.
2. Log in using your enterprise administrative credentials.
3. Locate the **Virtual Console** section on the BMC dashboard.
4. Launch the **HTML5 Virtual Console** (Preferred over Java for security and modern browser compatibility).

[PASTE IMAGE BLOCK HERE: Screenshot of iDRAC/iLO remote console UI dashboard showing system health and the 'Launch Virtual Console' button]

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4. Phase 2: Mounting RHEL ISO via Virtual Media

To boot the server, the RHEL 9.5 ISO must be mapped to the hardware as a virtual optical drive over the network.

1. In the active Virtual Console window, click the **Virtual Media** (or equivalent) menu.
2. Select **Map CD/DVD** or **Connect Virtual Media**.

3. Browse your local workstation files and select the rhel-baseos-9.5-x86_64-dvd.iso file.
4. Click **Map Device** and ensure the connection status shows as active/connected.

[PASTE IMAGE BLOCK HERE: Screenshot of the Virtual media mounting screen showing the RHEL 9.5 ISO successfully mapped as a virtual drive]

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5. Phase 3: Hardware & Pre-Boot Configuration

BIOS/UEFI Boot Configuration

1. Power cycle the server.
2. Enter the System Setup/BIOS during POST.
3. Ensure the **Boot Mode** is explicitly set to **UEFI**. Legacy BIOS is deprecated in RHEL 9.
4. **Secure Boot** should be configured according to your organization's security baseline (RHEL 9.5 fully supports Secure Boot).

RAID Configuration (Recommended)

1. Enter the RAID controller configuration utility (e.g., Dell PERC or HPE Smart Array).
2. Clear any foreign configurations if repurposing a server.
3. Create a **RAID 1 (Mirror)** Virtual Disk utilizing two physical drives for the OS volume.
4. Initialize the Virtual Disk.
5. Set the newly created Virtual Disk as the primary boot volume.

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6. Phase 4: Booting the RHEL Installer

1. Restart the server.
2. Press the key to enter the **One-Time Boot Menu** (e.g., F11 for Dell).
3. Select the **Virtual Optical Drive** (Virtual CD/DVD).
4. At the GRUB bootloader menu, use the arrow keys to select **Install Red Hat Enterprise Linux 9.5** and press Enter.

(Note: Do not select the "Test this media" option unless you are troubleshooting a corrupt ISO).

[PASTE IMAGE BLOCK HERE: Screenshot of the RHEL 9.5 GRUB Boot menu with 'Install Red Hat Enterprise Linux 9.5' highlighted]

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7. Phase 5: Anaconda Installer Walkthrough

7.1 Language Selection

Select the appropriate language and keyboard layout for your region (Default: English / United States) and click **Continue**.

[PASTE IMAGE BLOCK HERE: Screenshot of Anaconda Language selection screen]

7.2 Installation Summary

You will be presented with the main Installation Summary hub. All sections with a warning icon (⚠️) must be resolved before the installation can begin.

[PASTE IMAGE BLOCK HERE: Screenshot of the Anaconda Installation summary dashboard]

7.3 Disk Partitioning (LVM Recommended)

1. Click **Installation Destination**.
2. Select the target RAID virtual disk (e.g., sda).
3. Under Storage Configuration, select **Custom** and click **Done**.
4. In the Manual Partitioning screen, select **LVM** as the partition scheme.
5. Click **Click here to create them automatically** to generate a baseline template, then adjust to the following enterprise standard:
    - /boot: 1 GiB (Standard Partition, XFS)
    - /boot/efi: 600 MiB (EFI System Partition)
    - swap: Varies based on RAM (typically 4 GiB - 8 GiB)
    - / (root): 20 GiB+ (LVM, XFS)
    - /var: 10 GiB+ (LVM, XFS - highly recommended for isolated log growth)
    - /home: Remaining capacity or strictly allocated based on server role.
6. Click **Done** and **Accept Changes**.

[PASTE IMAGE BLOCK HERE: Screenshot of Disk partitioning screen showing an LVM layout with separate /var and /home mounts]

7.4 Network Configuration

1. Click **Network & Host Name**.
2. Enter the Fully Qualified Domain Name (FQDN) in the Host Name field (e.g., db-prod-01.internal.corp) and click **Apply**.
3. Select the primary management/production NIC and click **Configure**.
4. Navigate to the **IPv4 Settings** tab.
5. Change Method from DHCP to **Manual**. Add the static IP, Netmask, Gateway, and DNS servers.
6. Check **Require IPv4 addressing for this connection**.
7. Save, and toggle the Ethernet switch to **ON** in the upper right.

[PASTE IMAGE BLOCK HERE: Screenshot of Network config screen showing manual static IP entry and hostname applied]

7.5 Software Selection

1. Click **Software Selection**.
2. Under Base Environment, choose **Server with GUI** (if application requires X11/Wayland) or **Minimal Install** (Best practice for secure, headless servers).
3. Under Additional Software, select:
    - Standard
    - Hardware Monitoring Utilities
    - Guest Agents (if this bare metal server will host KVM VMs).
4. Click **Done**.

[PASTE IMAGE BLOCK HERE: Screenshot of Software selection screen highlighting 'Minimal Install' and 'Standard' add-ons]

7.6 User Settings

1. Click **Root Password**. Check **Lock root account** (forces the use of sudo for auditing purposes) OR set a highly complex administrative password. Click **Done**.
2. Click **User Creation**.

3. Create a primary administrative account (e.g., sysadmin).
4. Check **Make this user administrator** (adds user to the wheel group).
5. Set a strong password and click **Done**.

[PASTE IMAGE BLOCK HERE: Screenshot of User settings screen showing administrator user creation]

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8. Phase 6: Installation Execution & Validation

8.1 Installation Execution

Verify all warnings are cleared on the Installation Summary screen. Click **Begin Installation**.

[PASTE IMAGE BLOCK HERE: Screenshot of Installation progress bar screen displaying package deployment]

8.2 Reboot and Validation

1. Once the message "Complete!" is displayed, click **Reboot System**.
2. **CRITICAL STEP:** Immediately return to the BMC Virtual Console menu and **Disconnect/Unmap Virtual Media**. Failure to do so will cause the server to boot back into the installer.
3. Allow the server to boot past POST and into the OS.
4. Log in at the TTY prompt using the administrative credentials created in step 7.6.

[PASTE IMAGE BLOCK HERE: Screenshot of the First login prompt at the terminal interface]

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9. Enterprise Best Practices

- **Security Profiles (SCAP):** Use the Anaconda "Security Policy" module to apply Center for Internet Security (CIS) or DISA STIG baselines during the installation phase rather than retrofitting them later.
- **Time Synchronization:** Always configure Chrony (NTP) to sync with internal enterprise domain controllers or stratum 1/2 servers to prevent cryptographic and logging failures.
- **Subscription Management:** Register the system to your Red Hat Satellite server or RHSM immediately post-boot to pull the latest CVE patches before exposing the system to internal networks:

subscription-manager register --org="YOUR_ORG" --activationkey="YOUR_KEY"

- **Kdump Enablement:** Ensure Kdump remains enabled to capture kernel core dumps in the event of bare-metal hardware panics.

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10. Troubleshooting & Common Mistakes

Common Mistakes in Remote Installs

- **"The Boot Loop":** Forgetting to unmount the virtual media ISO before the final reboot is the most common cause of wasted time.
- **Insufficient /var Allocation:** Failing to isolate /var in a separate LVM volume. A runaway application log will fill the root partition and crash the entire system.
- **DHCP on Bare Metal:** Leaving NICs on DHCP without MAC address reservations. When the lease expires and changes, the server becomes orphaned.

Troubleshooting Matrix

Issue	Root Cause	Resolution
Virtual Console fails to load or drops connection	Network latency or strict firewall rules	Verify ports 443 and 5900 (VNC) are open to the BMC subnet. Lower virtual media bandwidth settings if on a high-latency WAN.
Installer hangs on "Starting installer..."	Slow virtual media read speeds	Instead of mapping a local file from your laptop, host the ISO on an NFS/CIFS share located in the same data center and mount it directly via the BMC interface.
No local disks detected in Anaconda	Missing RAID Driver / Uninitialized Array	Ensure RAID Virtual Disks were initialized in Phase 3. If using an unsupported controller, inject a Driver Update Disk (DUD) at the GRUB menu by pressing e and adding inst.dd.

Issue	Root Cause	Resolution
Network Unreachable during IP setup	Misconfigured switchport VLAN tagging	Verify with network engineering that the physical switch port connected to NIC1 is tagged for the correct server VLAN, not just the BMC VLAN.